

Technical Memo

TO: Technical Manager

FROM: Sadaf Sarbazi

RE: ESG Sentence Classification Model — Deployment Recommendation

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Summary

I developed a multiclass text classification model to categorize ESG-related sentences using the SASB-Aligned ESG Sentences dataset (6,460 sentences). The model assigns each sentence to one of six categories, including Environment, Human Capital, Social Capital, Governance and Non-ESG.

After evaluating multiple feature engineering approaches (CountVectorizer and TF-IDF) and machine learning algorithms (Naive Bayes and Logistic Regression), I recommend deploying Logistic Regression with TF-IDF features as the production model.

Why This Model

The primary evaluation metric is weighted F1 score, which balances precision and recall across the six classes while accounting for dataset imbalance.

The recommended model achieved:

- **Weighted F1 Score:** ~0.79
- **Precision:** ~0.79
- **Recall:** ~0.79

This performance exceeds the assignment requirement of $F1 \geq 0.70$ and outperformed Naive Bayes across most evaluation metrics. Logistic Regression also offers strong interpretability, enabling stakeholders to understand which words drive predictions. This transparency is valuable when deploying models for ESG-related analysis.

Key Limitation

The dataset is imbalanced, with the Non-ESG class representing a large portion of the data while categories, such as Business Model & Innovation, have significantly fewer examples. This imbalance can cause the model to occasionally misclassify minority classes or confuse related categories, such as Human Capital and Social Capital. To improve performance, I recommend expanding the training dataset with additional ESG-labeled examples and periodically retraining the model as new sustainability reporting language evolves.

Recommendation

Based on performance, interpretability and computational efficiency, I recommend deploying Logistic Regression with TF-IDF features for ESG sentence classification. While the model performs well on ESG-style reporting language, periodic retraining and additional labeled data will help improve robustness and maintain performance as ESG reporting practices evolve.